Autrin Hakimi

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PROFESSIONAL PROFILE

Software Systems Engineer specializing in **full-stack applications**, **AI/ML systems**, and **robotics automation**. Leads architecture, development, and deployment of production systems for semiconductor manufacturing, delivering a **seven-figure robotics** product and a high-impact ML-driven parts-matching tool. Published researcher with **ACM SIGSPATIAL** 2025 paper on **distributed deep learning** across **64 NVIDIA A100 GPUs**. Seeking to apply full-stack and AI expertise to drive innovative, high-value solutions in a forward-thinking engineering team.

TECHNICAL SKILLS

Languages: Python, C/C++, JavaScript, TypeScript, SQL, Bash, HTML, CSS, C#
Web & Backend: React, Python Flask, Tkinter, REST APIs, WebSockets, JavaScript/HTML/CSS
Cloud & DevOps: Docker, Git, CI/CD pipelines, PyInstaller/NSIS packaging, Linux/Unix, Microsoft Azure, Google Cloud, Snowflake
ML & Data: PyTorch, scikit-learn, Pandas, NumPy, distributed training, DGL, NetworkX
Databases: MySQL, SQL, SQLite, NoSQL, Snowflake
Automation & Control: Robot workflow orchestration, vision systems, HMI/GUI, ROS, OpenCV, robot controllers
Methodologies: Agile (Scrum/Kanban), test-driven development, cross-functional collaboration

PROFESSIONAL EXPERIENCE

ASM International Jan 2026 - Present
Software Systems Engineer *Phoenix, AZ*

- Own and deliver multiple **full-stack production systems** for semiconductor manufacturing-robotics platforms with **computer vision**, an ML-driven parts-matching tool, and operator tooling-using **Python, JavaScript, Flask, MySQL,** and **CI/CD pipelines**; the solutions streamlined production workflows and enabled rapid deployment to international sites
- Authored the majority of the codebase (**backend** and **frontend**) for the operator-facing collaborative robot application shipping on a seven-figure robotics product: **architected** the HMI and operator controls (guided workflows, manual jog, abort), engineered **multithreading** so live camera feeds and robot motion never freeze the UI, and built recovery and motion-safety logic including motion guards, safe return-to-home, **networking**, and **FTP**; **multi-unit orders worth tens of millions** from major customers ships on this product, and the application earned a quarterly team **award**
- Improved measurement accuracy and run reliability on a **robot/camera** conductance-measurement fixture, correcting motion, calibration, and geometry logic across many part configurations; built automated validation suites (140+ tests) verifying bore profile geometry and control logic off-hardware.
- Built a **full-stack ML-driven** parts matching system (Flask, JavaScript, SQL) that saves engineers hours of manual lookup, now the basis for a pending **patent** on which I am a named **inventor**; engineered **threading** and synchronization across robotics frontends to coordinate vision and control workflows
- Package and deploy applications to international manufacturing sites, and provide remote troubleshooting; this accelerated release cycles and lowered field-support response time
- Lead architecture and aligned with stakeholder to digitize a customer-facing qualification workflow, introducing Git version control for the team, which improved code traceability and reduced integration errors

Iowa State University, SwAPP Lab Aug 2024 - Nov 2025
Undergraduate AI Researcher *Remote*

- Co-authored 'HydroGAT' accepted at **ACM SIGSPATIAL 2025**: a **heterogeneous graph attention transformer** for flood prediction reaching NSE **0.97**; built **distributed training** across **64 NVIDIA A100 GPUs** on NERSC Perlmutter with **15× speedup**
- Engineered end-to-end data pipelines (**PyTorch, DGL, Pandas, NumPy**) for large-scale datasets; constructed heterogeneous graphs, built **AI models**, automated workflows via Bash scripting, cutting setup time **40%**
- Code: https://github.com/swapp-lab/HydroGAT Paper: https://dl.acm.org/doi/10.1145/3748636.3764172

BuilderTREND May 2024 - Aug 2024
Software Development Intern *Omaha, NE*

- Designed and shipped a React-based job proposal template, migrating legacy ASP pages to React and building full-stack features with JavaScript, TypeScript, and C# (.NET) during Kanban/Scrum sprints, which streamlined proposal creation and improved user experience

Alliant Energy May 2023 - May 2024
Data Science Intern *Remote*

- Developed predictive **ML models** (**Random Forest, SVM**) in **Python** and **SQL** analyzing electricity usage for **995K** customers in Snowflake; predicted EV ownership for customer segmentation, recognized by the Director of Data Analytics for impactful business insights.

PROJECTS

Defect Sense code

- Industrial **Visual Inspection**
- Built a local visual-inspection system using **PatchCore/Anomalib** and **FastAPI**, with **anomaly heatmaps**, **region localization**, and optional schema-constrained **VLM** reports.
- Evaluated on 63 MVTec AD bottle images with separate training and calibration sets: detected **48/48** defects with **0/15** false alarms, averaging **0.30 s/image** on an RTX 4070 Laptop GPU; added 145 GPU-independent tests.

Autonomous Robotic Chess Player code

- Built a fully **autonomous** system: piece recognition (**OpenCV**), **move calculation**, and **UR10e robot** arm control (**ROS, MoveIt**).

EDUCATION

B.S., Computer Science **Aug 2021 - May 2025**
Iowa State University *Ames, IA*

- **GPA:** 3.70/4.0
- **Achievements:** Magna Cum Laude, Dean's List